

Grayscale-to-Color Translation for Animal Images with CGANs and RPNs

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Abstract

Animal image colorization is a challenging image-to-image translation task due to the diversity of textures, fur patterns, and semantic ambiguity across species. While cGAN-based approaches with U-Net generators and PatchGAN discriminators perform well, standard PatchGANs treat all image patches uniformly and do not explicitly incorporate semantic structure. This work investigates whether introducing semantic guidance into the discriminator improves animal colorization. We build on the Pix2Pix-based Colorization model, which maps the luminance L channel to the ab channels in LAB space, preserving structural information while learning chromatic prediction. To incorporate semantics, we use Region Proposal Network (RPN) guidance to construct spatial masks that reweight PatchGAN losses based on region importance. We evaluate three models: a baseline Colorization model, and two RPN-weighted variants (one initialized from pretrained weights and one trained from scratch). Quantitatively, the ResNet classifier shows that the pretrained RPN-weighted model improves accuracy by 2.6% over the baseline, while the scratch-trained RPN model improves by 1%. Qualitative results further indicate improved object-level consistency and color coherence under semantic weighting.

1. Introduction

Image colorization is the task of generating realistic color representations from grayscale images while preserving the underlying semantic and structural contents of the scene. In the context of animal imagery, this problem is particularly challenging due to the diversity of fur patterns texture, and natural environments across species. Accurate animal colorization may be useful for restoring archival wildlife photography, enhancing ecological datasets, and improving downstream computer vision applications that benefit from color information. While modern deep learning methods have achieved strong colorization performance, much of the existing work focuses primarily on human-centered or man-made subjects such as faces and urban scenes. Comparatively less attention has been given to animal-specific col-

orization, where semantic ambiguity and fine-grained texture variation make realistic color prediction more difficult.

Conditional Generative Adversarial Networks (cGANs) [2] with U-Net generators and PatchGAN discriminators have become a widely adopted model for image-to-image translation tasks, including image colorization. In this architecture, the U-Net generator learns a mapping from grayscale inputs to color outputs, while the PatchGAN discriminator evaluates the realism of local image regions to enforce fine-grained texture and color consistency. Traditionally, the discriminator operates on uniformly samples patches distributed across the image, treating all spatial regions with equal importance during adversarial training. This motivates the central research question of this work: can the PatchGAN discriminator be improved to learn higher-level semantic structure in addition to enforcing local texture realism?

Experiments were conducted using Animals-10 by Carrado Alessio, which contains approximately 28,000 images spanning 10 animal classes: butterflies, cats, cows, chickens, dogs, elephants, horses, sheep, spiders, and squirrels. The images were collected from Google and therefore exhibit substantial variation in quality, resolution, aspect ratio, and scene composition. Due to computational limitations associated with the free tier of Google Colab, a random subset of 3,000 images was selected for experimentation. The dataset was partitioned using a 80:10:10 training, validation, and test split. To enforce spatial uniformity across samples, images were centered cropped to a fixed 128×128 resolution, which also served as a lightweight form of data augmentation by improving robustness to varying image dimensions and compositions. Corresponding grayscale inputs were generated by averaging across the color channels of each image.

This work makes several contributions toward improving semantic awareness in GAN-based animal colorization. First, we employ the Colorization model, which extends the existing Pix2Pix framework by specializing the image-to-image translation task for colorization. The model reformulates colorization as a mapping from the luminance L channel to the ab color channels in LAB color space. This representation decouples grayscale structure from chromatic in-

formation, allowing the model to preserve structural content through the known L channel while focusing exclusively on predicting natural color information in the ab channels. Second, we incorporate Region Proposal Network (RPN) guidance into the PatchGAN discriminator to introduce semantic awareness during adversarial training. Rather than treating all uniformly sampled patches equally, we construct an RPN-based weighting mask that estimates the semantic importance of different image regions according to their overlap with region proposals. The adversarial loss associated with individual discriminator patches is then reweighted using this mask, encouraging the discriminator to place greater emphasis on semantically meaningful regions during optimization. Finally, we evaluate three distinct training configurations: a baseline Colorization model trained from random initialization, a Colorization model using the RPN-weighted discriminator loss initialized from pretrained baseline weights, and a Colorization model using the same RPN-weighted loss trained entirely from random initialization.

Model performance was evaluated using both quantitative and qualitative analysis. Quantitatively, we measured downstream classification performance using state-of-the-art image classification architectures to assess how effectively semantic information was preserved in generalized colorizations. With respect to the ResNet classifier, the Colorization model using the RPN-weighted discriminator loss with pretrained initialization improved classification accuracy by 2.6% over the baseline model, while the RPN-weighted model trained from random initialization improved accuracy by 1%. Qualitatively, we conducted visual comparisons between generated outputs and ground truth image to analyze perceptual realism and identify weaknesses in the learned colorization mappings. Comparisons between the baseline model and the semantically aware RPN-weighted models also allowed us to examine how semantic guidance influence object-level consistency and color coherence. The Colorization model trained with the RPN-weighted discriminator looked very similar to the original Colorization model. The RPN-weighted model trained from random initialization had less mix up of background coloring and subject coloring, a natural result of focusing on the subject and not the background.

2. Related work

Isola et al. [2] first applied the modern conditional GAN framework to image-to-image translation. GANs are comprised of two main components: a generator and a discriminator. Working on paired images, the generator seeks to fool the discriminator by generating a synthetic image that resembles the true output. On the other hand, the discriminator tries to identify the true image as real and the synthetic image as fake. Using GANs enables the discriminator to dy-

namically learn a loss function for an arbitrary translation task. As a result, GANs serve as general-purpose models for image-to-image translation. Since the loss function is learned by the discriminator, it provides a meaningful assessment of the realism of the generated image. This adversarial loss captures local consistency, while an additional L1 loss term promotes overall reconstruction accuracy.

In [5], the researchers introduce the CycleGAN architecture for settings without paired data. The key contribution of this work is the cycle-consistency loss, which enforces the principle that reversing a transformed image should reconstruct the original input image. In another prominent work, the researchers resolved the stochasticity issue in the image generation process. The authors incorporate a latent regression loss that measures how effectively the latent noise vector can be recovered from the generated outputs image. As a result, diverse outputs can be generated from the same input.

Huang et al. [1] deconstructs images into domain-invariant content and domain-specific style. When creating the output image, a domain-specific style is used to enrich the domain-invariant content. In this way, the model preserves the high level structure of the input image. At the same time, randomness is injected in an explicit manner so the base image can be transformed into unique output images. Another framework discussed in [3] tries to incorporate fine-grained information about an input image. The authors observe that standard normalization layers can wash out semantic structure. In the paper, the authors propose spatially-adaptive normalization, which normalizes activations using spatially varying parameters derived from segmentation maps. This new approach significantly improves image quality and controllability.

With L1 loss, GANs default to average values when uncertain about a pixel region resulting in a blurry image. Pix2PixHD addresses this problem by using a coarse-to-fine generator architecture and multiple discriminators operating at different image scales. This allows the model to capture both global structure and local detail.

While prior work has made significant progress in image-to-image translation, existing methods largely rely on pixel-level reconstruction losses and spatially uniform adversarial objectives. As a result, they often fail to enforce semantic consistency, producing outputs that are visually realistic but contain object-level inaccuracies. Furthermore, approaches to modeling diversity introduce stochasticity without explicitly constraining it to semantically meaningful variations. These limitations highlight the need for methods that incorporate object-level supervision and region-aware adversarial learning to better align generated outputs with the underlying structure of the scene.

3. Method¹

We adapt the cGAN architecture described in [2] to animal colorization. The generator maps a grayscale image to a colored image. In [2], the cGAN generator also accepts a noise vector z to add stochasticity to the outputs. However, the approach was unsuccessful as L1 loss discouraged creative outputs far from the ground truth. Instead the authors relied on dropout to apply fine-grained Gaussian perturbations. We prioritized accurate outputs and therefore did not introduce randomness into the generator.

The discriminator takes as input a grayscale and colored image pair and outputs the probability that the colored image is real given the grayscale image. In [2], the images are divided into uniform $N \times N$ patches and are fed into the PatchGAN architecture. By focusing on small regions in the image, the discriminator would catch low-level inconsistencies and blurring caused by L1 loss.

The loss function for the model is the sum of cGAN loss which incentives realistic outputs and L1 loss which encourages overall accuracy. The discriminator seeks to maximize this loss and the generator aims to minimize it. cGAN loss is given by

$$\mathcal{L}_{\text{cGAN}}(G, D) = \mathbb{E}_{x,y}[\log D(x, y)] + \mathbb{E}_y[\log(1 - D(x, G(x)))]$$

where x is the grayscale image, y is the original colorized image, G is the generator, and D is the discriminator.

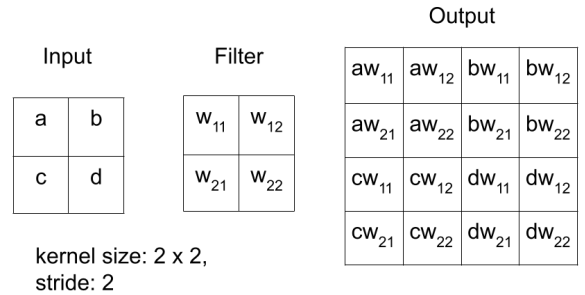
Specifically, the discriminator aims to maximize the probability it assigns to the true pair while minimizing the probability it assigns to the generated pair. On the other hand, the generator seeks to maximize the probability that the discriminator assigns to the generated pair. The L1 loss $\mathcal{L}_{\text{L1}}(G) = \mathbb{E}_{x,y}[||y - G(x)||_1]$ compares the output of the generator to the true colored image. L1 loss is used instead of L2 loss, as L2 loss penalized larger deviations from the ground truth more heavily, often resulting in blurred outputs. Expectations with respect to all grayscale images x and colored images y can be approximated using an empirical average over the dataset. At a high level, the objective function characterizes a distance metric between the data distribution p_{data} and the generator’s distribution p_G .

Isola et al. [2] developed a general purpose Pix2Pix model which captures RGB to RGB translation. Pix2Pix was applied to a series of tasks such as day to night and edges to photo translation. We adapted the author’s PyTorch implementation of the Colorization model which extends Pix2Pix. To support grayscale to color translation, the Colorization model maps a grayscale representation to LAB color space. The L channel captures structural and intensity information such as edges, textures, and object boundaries. The a channel measures the green to red balance and the

b channel measures the blue to yellow balance. Adopting the LAB color space decouples the luminance of an image which can be directly represented by a grayscale image with the chromatic information. When the model is given a grayscale image, the L channel is already fully defined, allowing it to concentrate solely on learning and predicting the complementary ab color channels.

The generator is implemented using a classic encoder-decoder architecture first leveraged for semantic segmentation tasks. The encoder builds a dense representation of the input and creates a bottleneck preventing the model from learning the trivial mapping between pixels. The decoder applies upsampling layers to increase the spatial dimension to the size of the output. Since the decoder is working with a condensed representation, it fails to incorporate local relationships in the image. The U-Net architecture remedies this issue by providing residual connections between symmetric encoder-decoder layers. In this way, a decoder layer can upsample using a rich representation from the previous layer and a detailed representation from the encoder.

We employed the U-Net 128 generator architecture for 128×128 images. Encoder layers begin with a 128×128 input and progressively reduce the spatial dimensions by a factor of two over seven layers. The number of channels begins at 64 in the first layer and doubles at each subsequent layer, up to a maximum of 512 channels. Each encoder block consists of a convolution, batch normalization, and ReLU activation. After a single bottleneck layer, the decoder performs transposed convolutions to upsample the feature maps, concatenates them with corresponding encoder features via skip connections, and applies convolutions to fuse the combined representations into a coherent reconstruction. In U-Net 128, transposed convolutions were implemented using a 2×2 kernel with stride 2. These settings increase the spatial dimension by a factor of two by interpolating each input pixel using the convolutional kernel. The upsampled image matches the dimensionality of the encoder feature and produce a image with twice the channels when concatenated. The final convolution layer reduces the number of channels by a factor of two to be compatible with the subsequent decoder block.



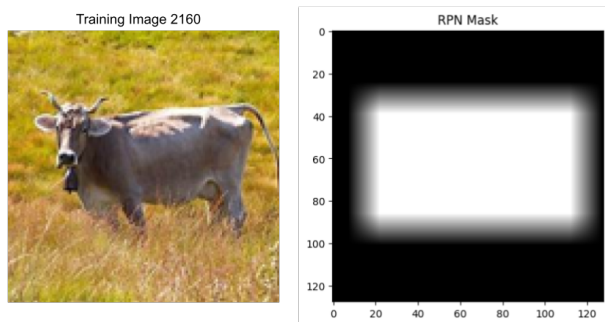
We use the PatchGAN discriminator with 70×70 patches, constructed using multiple convolutional layers un-

¹Codebase (Shared Google Drive): https://drive.google.com/drive/u/1/folders/0AHs5K9_BlSkQUk9PVA

til the receptive field matches the patch size. The output of the discriminator is a 2D map that assigns a score to each patch. Loss is computed independently for each patch using the binary cross-entropy formulation described above, based on whether the input image is real or generated. Finally, the patch-wise losses are aggregated using an empirical average.

Despite the benefits, the PatchGAN discriminator is applied to regions uniformly across the scene. The approach fails to consider the object-level features and thus cannot describe the semantics of the image. To this end, we leverage a pretrained region proposal network to enable the PatchGAN discriminator to identify semantically meaningful patches. Historically, region proposals were conducted using unsupervised learning algorithms like K -Means clustering. In Fast R-CNN, the computational overhead associated with region proposals acted as a bottleneck to performance. Ren et al. [4] introduced Region Proposal Networks (RPNs) to generate region proposals for the Fast R-CNN architecture. Working in feature space, multiple anchor boxes are created around each pixel at different scales and aspect ratios. An RPN is a CNN that predicts whether an anchor box represents an object. RPNs also determine the scale and shift transformations required to land at a region proposal. As a result, RPNs induce two additional loss terms for evaluating the entire network. The first assesses the region proposal prediction and the second measures the accuracy of the transformation.

To integrate RPNs, we relied on the pretrained Faster R-CNN model to propose regions of interest for all training images. For each image, we discarded region proposals with scores below 0.7 and converted the remaining proposals into an RPN mask. The RPN mask is a 2D map indicating whether each pixel belongs to a region proposal.



During training, we used the RPN masks to modulate the discriminator loss. We resized the RPN mask for each evaluated image to match the output resolution of the PatchGAN. Next, we clamped the RPN values to the range $[0.2, 1.0]$ to prevent zeroing out gradients completely. We then computed the average binary cross-entropy loss over all patches, weighed by the RPN mask. This RPN-weighted discriminator loss imposes a larger penalty on the discrim-

inator for misclassifying semantically meaningful patches. As a result, the discriminator focuses on identifying objects in the scene that exhibit the greatest variation across images. In addition to enforcing low-level consistency, a more semantically aware PatchGAN can improve classification performance.

To support efficient training, we precomputed RPN masks for all training images a priori. For each epoch, the same RPN masks were reused to compute the loss and optimize the model parameters. We did not compute RPN masks for the validation and test sets. During validation, performance was evaluated using the empirical average of the patch losses rather than a weighted average based on the RPN masks. At test time, we solely evaluated the images produced by the generator.

4. Results

We trained three independent models to perform animal colorization. The first follows the base Colorization model implemented in [2], which builds upon the Pix2Pix architecture by mapping the L channel to the ab channels in the LAB color space. The model is initialized with random weights. The second model extends the Colorization architecture by incorporating an RPN-weighted discriminator loss. It is fine-tuned using pretrained weights from the baseline model. The third model employs the colorization architecture with the RPN-weighted discriminator loss, but is initialized with randomly generated weights.

We used a set of hyperparameters chosen to promote stable adversarial optimization during training. The base Colorization model was trained with a U-Net 128 generator on 128×128 images, balancing spatial detail preservation with computational efficiency. Both the load size and crop size were fixed at 128 with no additional preprocessing to maintain spatial consistency between inputs and targets. Training was performed with a batch size of 1, following the standard Pix2Pix formulation for image-to-image translation and helps maintain diversity across gradient updates. The model was trained for 50 epochs at a constant learning rate followed by 50 epochs of linear learning rate decay to encourage stable convergence during later training stages. Checkpoints were saved every 5 epochs, and losses were logged every 100 iterations to monitor optimization behavior throughout training.

During training, four primary statistics were logged to monitor optimization behavior: G_{GAN} , G_{L1} , D_{real} , and D_{fake} . The generator adversarial loss G_{GAN} measures the binary cross-entropy loss between discriminator predictions on generated samples and the real target label, quantifying how successfully the generator fools the discriminator; lower values generally indicate more realistic generated images. The reconstruction loss G_{L1} measures the mean absolute error between the generated and ground truth ab color

channels in LAB color space, encouraging accurate color reconstruction and spatial consistency; lower values correspond to stronger pixel-level agreement. The discriminator real loss D_{real} measures the binary cross-entropy loss on real images pairs, where lower values indicate stronger recognition of authentic samples. Similarly, the discriminator fake loss D_{fake} measures the binary cross-entropy loss on generated image pairs, with lower values indicating improved detection of synthesized outputs.

In the second model, most hyperparameters were kept consistent with the baseline configuration to ensure a controlled comparison, including the U-Net 128 generator architecture, 128×128 image resolution, batch size of 1, and identical preprocessing settings. The primary additions were the use of RPN masks and a dedicated mask directory, to enable RPN-weighted discriminator loss. Unlike the baseline model, this routine was initialized using pre-trained weights from the previously trained Colorization model, allowing optimization to begin from an already stable solution rather than random initialization. The learning rate was reduced to prevent large parameter updates during fine-tuning. Training was performed for 20 epochs without learning rate decay. For the third model, which applied the RPN-weighted discriminator loss while training from random initialization, the same hyperparameter configuration was maintained to ensure that performance difference could be attributed to the training strategy rather than changes in optimization settings.

As mentioned in [2], evaluating the results of these images is an open problem. We can compare a per-pixel difference, as we do for the loss function. This is an easy numerical way to evaluate, although it is not a great way for evaluating the holistic result. A good metric for evaluating the quality of our colored results is to test whether or not humans can tell the difference between our generated images and real images. Moreover, simply looking at the result and seeing if it makes sense is a good way to get a rough estimate of the quality of the images. We will use this method to evaluate some of the results, but due to time and resource constraints, we cannot use this method of evaluation extensively. As another alternative, we can use pretrained models to classify both our generated images and the ground truth. If the accuracy of the pretrained models are similar in both cases, then our generated results are semantically similar to the ground truth. This metric similar to using human differentiation, but more feasible to employ over large quantities of data.

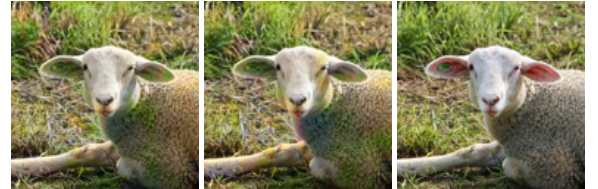
Specifically, we used GoogLeNet and ResNet50, two common image classification models, to evaluate the performance of our model. As both evaluation models work best with images of size 224×224 , we scaled our 128×128 images to 256×256 , then cropped to only take the center 224×224 pixels of the result.

We ran the evaluation models on 299 test images. For each test image, we compared the classification of colorized test image with the classification of the generated image. A model that generates more realistic colorizations should produce outputs that are recognizable as the same animal as in the original image. We test three different models: (1) the basic Colorization model as implemented in [2] (2) the Colorization model Below are the results given as the proportion of test images where the generated image was classified the same as the original image.

	GoogLeNet	ResNet50
Colorization	183/299	195/299
With RPN	186/299	200/299
With RPN (Scratch)	181/299	197/299

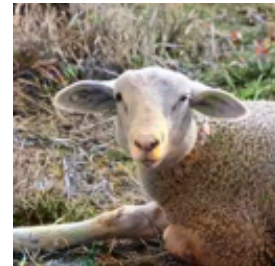
Although the differences were slight, we found that using an RPN with the PatchGAN consistently performed slightly better on the quantitative results than the default PatchGAN. The model trained from scratch performed about as well as the Colorization model.

Visually, we found the Colorization model with and without the RPN to look similar. Through visual inspection, we found that both were prone to background bleed-through, where parts of the animal would take on colors from the background. such as with the image of the sheep below. From left to right, in order, we have the image generated by the Colorization model, the image generated by colorization with RPN, and the original image.



In the generated image, the sheep takes on some of the hues of the grass, and its wool has a yellower tint reminiscent of the dry grass in the background.

The model trained from scratch, although not quantitatively better, does perform qualitatively better in some aspects. It suffers from less background bleed-through, as can be seen by the generated image of the same sheep below.

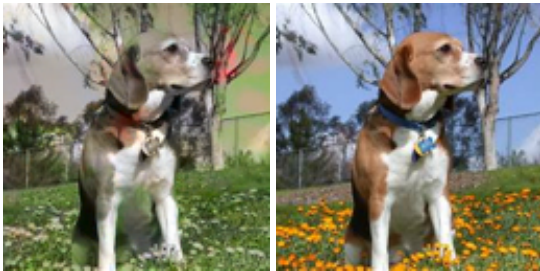


In this generated image, there is practically no green on the sheep at all. Nevertheless, all three models have general tendency to have less contrast in their generated images than in the ground truth. Below are images of the same squirrel. Top left is generated from the Colorization model, and top right is from the Colorization model with an RPN. Bottom left is the generated from the model trained from scratch, and bottom right is the ground truth.



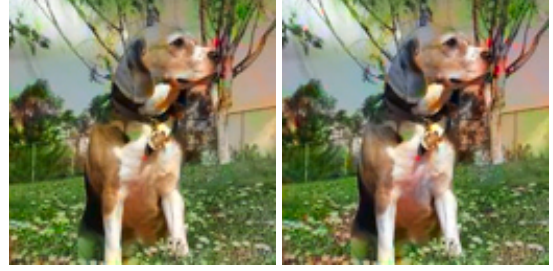
All the models produced fairly reasonable results, although the squirrel in the model trained from scratch looks more like a chipmunk. The models all have a similar color theme and one main flavor of color.

The model trained from scratch does not always perform well. In some images, the model leaves some areas uncolored or less colored, as can be seen below. To the left we have the image generated by the model trained from scratch, and on the right is the ground truth.



In the generated image of the dog, much of the dog, sky and flowers are gray, although the grass is the proper shade of green. In addition, part of the sky is green, and there is a patch of brighter red.

The Colorization model (with and without RPN) perform better in this aspect as can be seen below.



In both generated images, the dog is the shaded the correct color for hte most part, although the back of its head is still somewhat gray, and its chest is erroneously brown. Slightly more color is applied to the flowers near the bottom, and the left part of the sky is a grayish blue. Nevertheless, the sky on the right is still pink. We also find that the sky around the green branches are green. Likely, the models learned that green leaves commonly occur around tree branches, and applied it here as well, even though the leaves here are sparse.

Overall, as expected, the Colorization model shares many similarities, both positive and negative, with the version of it with an RPN. The model trained from scratch performs inconsistently compared to the other two models. It lacks some of the negative features of the models such as background bleed-through, but also colors less vividly at times.

5. Conclusion

This section discusses the limitations of our research and outlines potential directions for future work. One limitation of this work is the inherent instability associated with adversarial training in GAN-based frameworks. Because the generator and discriminator are optimized simultaneously, training can become unstable if one network learns significantly faster than the other and begins to dominate the optimization process. In particular, an overly strong discriminator may prevent the generator from learning meaningful gradients, while an overly strong generator can reduce the discriminator's effectiveness. Additionally, adversarial learning introduces uncertainty in determining when the model has truly converged, since generator and discriminator losses may continue fluctuating even when qualitative outputs appear visually stable. Finally, while the generator can produce plausible colorizations, interpreting the internal representations and understanding how semantic structure is associated with specific color predictions remains difficult, limiting the interpretability of the learning mapping.

Future work could explore replacing the adversarial framework with a Variational Autoencoder (VAE) formulation for conditional image colorization. In this setting the colorization process could be modeled through a joint probability distribution over grayscale images, latent semantic

representations, and colored outputs. The grayscale image would condition a latent variable capturing high-level structural and semantic information, which could then be used to generate the final colorization. Unlike GANs, which optimize through a competitive game between two networks, VAEs are trained by maximizing a variational lower bound on the data likelihood using stochastic variational inference. This produces a single, well-defined optimization objective, generally leading to more stable dynamics and clearer convergence behavior compare to adversarial optimization. Additionally, the latent representation learned by a VAE may provide improved interpretability, as the latent dimensions can capture meaningful modes of variation across grayscale images such as texture, object structure, or plausible color semantics. Conditioning on a grayscale image produces a distribution over these latent features, allowing the model to sample diverse but coherent semantic representations before generating a corresponding colorization in a principled probabilistic manner.

Future work could also explore diffusion models for conditional image colorization. During training, the model would learn a conditional denoising process by taking a grayscale image together with a progressively noised version of the ground truth color channels and learning to reconstruct the original color information. In this formulation, the grayscale image provides structural guidance throughout the denoising process, allowing the network to learn a conditional probability distribution over plausible colorizations. Unlike GANs, diffusion models optimize a direct reconstruction-based objective rather than a competitive adversarial objective, generally leading to more stable optimization dynamics and more predictable convergence behavior. During inference, the model would generate a color image by starting from random noise in the color channels, and iteratively refining the sample while conditioning on the grayscale input at every denoising step. This sequential generation process improves interpretability, as intermediate denoising stages reveal how semantic structure and color information are progressively organized into a coherent final colorization in a principled probabilistic manner.

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